EAST CRAZIES EA SOILS SCOPE OF WORK

Date: 2026-05-07

Project: East Crazy Inspiration Divide Land Exchange Environmental Assessment
Forest: Custer Gallatin National Forest
District: Bozeman Ranger District
Resource area: Soils

Prepare the soils work needed to support the East Crazies EA record. The work must determine whether the existing EA and Plan Consistency Table soil conclusions are still supported, or whether the Forest Service needs revised soils analysis, mitigation, monitoring, or EA text.

Focus the scope on the soil issues created by the proposed action:

- Sweet Trunk Trail No. 274, including the new trail alignment and drainage design.
- Inspiration Divide Trail No. 8 reroute, including the removed stream/wetland crossing.
- Big Timber Canyon Trailhead improvements.
- Stream crossings, wetland/riparian approaches, wet soils, hydric soils, erosion, and sediment delivery.
- Parcel restrictions, wetland covenants, access/easements, grazing-related improvements, and any acquired-parcel soil management needs.

The expected product is a Soils Specialist Report unless the Forest Service soils reviewer approves a short no-issue or incorporated-analysis memo. Either product must address the fieldwork, analysis, and deliverables below.

This is a planning and contracting support SOW. It is not a final agency decision, legal sufficiency review, or NEPA compliance determination.

FIELDWORK

1. Pre-Field Review

Before mobilization, review:

- final EA, proposed action, alternatives, maps, Plan Consistency Table, Roads/Trails/Access Report, Wetlands Report, and parcel restriction documents;
- Sweet Trunk Trail alignment, Inspiration Divide Trail reroute, Big Timber Canyon Trailhead footprint, easements, access routes, stream crossings, wetlands, riparian areas, parcel boundaries, and staging/restoration areas;

- NRCS Web Soil Survey/SSURGO soil map units, slope, erosion hazard, drainage, hydric soil ratings, compaction/rutting limits, and revegetation limitations;
- DEM/slope data, NHD, NWI, MTNHP wetland/riparian data, aerial imagery, route layers, grazing allotment/improvement layers, and any available landslide or mass-wasting data;
- Custer Gallatin soil, watershed, road/trail, riparian, recreation, invasive species, and backcountry/recommended-wilderness plan components that affect soil work.

2. Fieldwork Plan

Submit a fieldwork plan for Forest Service review before fieldwork starts. Include:

- field questions and decision points;
- activity-area definitions for trail, trailhead, access, wetland/riparian, grazing-improvement, and parcel-management features;
- maps showing soil map units, slopes, proposed review areas, trail/road alignments, crossings, wetlands, riparian areas, and parcel boundaries;
- proposed observation points, transects, meander routes, representative soil checks, and photo points;
- method for documenting soil disturbance for linear trail features;
- method for deciding whether FW-STD-SOIL-01 applies, does not apply, or needs a linear-feature explanation;
- GPS/photo naming, field forms, data transfer format, access needs, safety constraints, and coordination needs.

3. Field Review Areas

Review the following where access allows:

- Sweet Trunk Trail alignment.
- Inspiration Divide Trail reroute.
- Big Timber Canyon Trailhead improvement area.
- Stream crossings, wetland/riparian approaches, seeps/springs, slope breaks, toe slopes, drainage paths, and sediment-routing areas.
- Federal parcels leaving ownership and non-federal parcels entering ownership where parcel restrictions or future NFS management create soil questions.
- Existing trails, roads, route drainage, grazing improvements, compacted areas, eroding areas, disturbed areas, and revegetation/restoration areas.

4. Field Methods

Use field methods sufficient to support report-ready conclusions:

- verify representative soil map units, landforms, slope classes, drainage, surface texture, coarse fragments, hydric/wet soil indicators, and erosion susceptibility;
- classify visible soil disturbance, including compaction, rutting, displacement, puddling, surface erosion, organic-matter loss, and mass movement;
- document representative conditions with GPS points/tracks/polygons and photos;
- use shovel checks, probes, or soil pits where needed for hydric indicators, rooting depth, restrictive layers, wetness, compaction, or restoration feasibility;
- check whether trail and trailhead design can hydrologically disconnect drainage from waterbodies and riparian management zones;
- identify locations needing drainage controls, crossing hardening, wet-weather restrictions, erosion control, revegetation, decompaction, weed prevention, or monitoring.

Detailed lab sampling is not required unless a specific soil question cannot be resolved through soil survey, visual methods, field checks, and professional judgment.

5. Minimum Field Record

For each activity area or representative field check, record:

- date, observer, weather, soil moisture, and access limits;
- activity-area ID, action element, GPS identifier, and photo IDs;
- soil map unit and field-verification notes;
- slope, landform, drainage position, hydric/wet soil indicators, and sediment-connectivity notes;
- existing disturbance type and soil-disturbance class;
- expected disturbance footprint, length, width, acreage, crossing count, or other useful measure;
- BMP, design feature, mitigation, monitoring, data gap, or revisit need.

Fieldwork should occur during snow-free conditions when the soil surface can be observed. Recommend a wet-season or post-storm revisit if runoff, drainage, wetland connectivity, erosion, or sediment delivery cannot be evaluated during the main field visit.

ANALYSIS

1. Existing Condition

Describe the soil setting for the affected action areas:

- soil map units, slopes, parent materials, drainage, erosion hazard, compaction/rutting limits, revegetation limits, hydric soil indicators, and sensitive soil attributes;
- existing roads, trails, trailheads, stream crossings, wetland/riparian areas, grazing improvements, disturbed areas, and route drainage;
- existing soil disturbance and sediment connectivity;

- soil conditions on disposal and acquisition parcels where relevant to the land exchange decision;
- data gaps and confidence limits.

2. Effects

Analyze no action, proposed action, Alternative 1 if retained, and any other Forest Service-provided alternative. Address:

- soil disturbance from trail construction, trail reroute, trailhead work, crossings, access, staging, restoration, and parcel-management changes;
- whether trail work near or across waterbodies could deliver sediment, and whether design features keep the magnitude and duration low;
- whether the Trail No. 8 reroute reduces soil and sediment effects by removing the existing stream/wetland crossing;
- whether wetland covenants, no-subdivision restrictions, no-mineral-development restrictions, and retained wetland parcels reduce foreseeable soil disturbance;
- whether acquired parcels create new soil-management needs for NFS management;
- whether trailhead resurfacing, parking, toilet facilities, or kiosk construction require erosion, compaction, drainage, topsoil, or revegetation controls;
- whether grazing improvements, grazing use, hazardous materials, minerals, vegetation, invasive species, restoration, or carbon assumptions create soil handling or mitigation needs.

Quantify disturbance where useful. For linear features, use length, width, crossing count, sensitive soil crossings, wetland/riparian approaches, drainage features, and representative disturbance class instead of forcing all effects into an acreage metric.

3. Forest Plan Consistency

Evaluate and document consistency with:

- FW-DC-SOIL-01, FW-DC-SOIL-02, and FW-DC-SOIL-03;
- FW-STD-SOIL-01, including the rationale for applying or screening out the standard;
- FW-GDL-SOIL-01 through FW-GDL-SOIL-08;
- related road/trail direction, especially hydrologic disconnection, mass-wasting avoidance, and hardened stream crossings;
- invasive species, revegetation, riparian, wetland, aquatic, recreation, backcountry, and recommended-wilderness components that affect soil or sediment analysis.

The report must state where it agrees with, narrows, or revises the existing Plan Consistency Table.

4. Mitigation, BMPs, And Monitoring

Prepare a mitigation/BMP/monitoring matrix with action element, location trigger, soil concern, required measure, responsible party, timing, and verification method. Include, as applicable:

- trail drainage spacing, rolling dips, grade reversals, waterbars, cross drains, armoring, outsloping, and sediment filtering;
- hydrologic disconnection from waterbodies and riparian management zones;
- stream-crossing hardening for trail approaches, streambeds, banks, and approaches;
- avoidance or design modification for wet soils, hydric soils, steep slopes, unstable areas, high mass-wasting potential, and sensitive soil map units;
- wet-weather restrictions, erosion control, decompaction, topsoil/duff salvage, revegetation, mulch, weed-free seed, weed prevention, restoration, and post-construction photo-point monitoring.

5. Cumulative Effects And Uncertainty

Evaluate cumulative soil effects only where there is a useful cause-and-effect relationship. Consider existing and proposed routes, trailheads, unauthorized routes, route drainage, grazing, wetland or riparian disturbance, foreseeable trail or road work, private land development risk after conveyance, and vegetation/restoration actions that could affect ground cover or erosion.

Clearly identify unresolved field limitations, assumptions, and Forest Service decisions needed before soil conclusions can be used in the EA, decision record, collection agreement, trail design package, or implementation file.

DELIVERABLES

1. Kickoff And Data Request Log

Provide kickoff notes, received and missing data, access constraints, source documents reviewed, stop conditions, and Forest Service decisions needed before fieldwork.

Acceptance criteria:

- identifies missing GIS, trail design, parcel, wetland, route, or Forest Plan data;
- identifies whether the work will proceed as a standalone soils report or no-issue memo;
- identifies field-season and access constraints.

2. Fieldwork Plan

Provide field objectives, activity-area definitions, field maps, observation methods, field forms, data conventions, schedule, safety/access plan, and coordination needs.

Acceptance criteria:

- covers Sweet Trunk Trail, Inspiration Divide reroute, Big Timber Canyon Trailhead, crossings, wetland/riparian interfaces, and parcel restrictions;
- explains how soil disturbance will be documented for linear trail features;
- identifies whether wet-season or post-storm verification may be needed.

3. Field Data Package

Provide GPS data, completed field forms, photo log, soil-check notes, soil map unit verification, wetland/riparian and sediment-connectivity notes, and a short field findings memo.

Acceptance criteria:

- every material field conclusion is traceable to location, date, observer, photo, and activity area;
- sensitive soil, hydric/wet soil, erosion, compaction, rutting, displacement, drainage, and sediment concerns are mapped or explicitly screened out;
- data are suitable for Custer Gallatin GIS and ID-team use.

4. Analysis Tables And Map Atlas

Provide:

- soil map unit and activity-area crosswalk;
- existing and expected disturbance table;
- action-element by soil concern matrix;
- stream/wetland crossing and sediment-connectivity table;
- mitigation/BMP/monitoring matrix;
- Forest Plan consistency matrix;
- data-gap and uncertainty table;
- project, soil-analysis area, soil map unit, slope/erosion hazard, wetland/riparian, road/trail, observation, and mitigation maps.

Acceptance criteria:

- maps include scale, north arrow, coordinate system, data date, project boundary, activity-area IDs, and source attribution;
- tables distinguish no action, proposed action, Alternative 1, and any other analyzed alternative;
- linear-feature effects use useful measures and do not rely on misleading acreage-only metrics.

5. Draft Soils Specialist Report Or No-Issue Memo

Provide a draft report or approved no-issue memo with affected environment, methods, spatial and temporal scale, resource indicators, direct and indirect effects, cumulative effects, mitigation/BMPs/monitoring, Forest Plan consistency, references, and appendices.

Acceptance criteria:

- directly addresses the final EA, Plan Consistency Table, Roads/Trails/Access Report, and Wetlands Report;
- states whether FW-STD-SOIL-01 applies or is screened out for trail work;
- ties conclusions to field evidence, GIS evidence, source documents, or specialist judgment.

6. EA-Ready Soils Text

Provide concise text for the EA or appendix covering affected environment, environmental consequences, mitigation/design criteria, Forest Plan consistency, and cumulative effects if needed.

Acceptance criteria:

- suitable for public disclosure;
- references the soils report or no-issue memo instead of duplicating all technical detail;
- identifies any design criteria or monitoring that must carry into the decision or implementation file.

7. Final Report And Data Package

Provide final report or memo, final PDF, editable source file, comment-response log, final tables, map atlas, GIS layers or geodatabase/geopackage, photo log, field forms, source data inventory, metadata/readme, and data limitations memo.

Acceptance criteria:

- resolves Forest Service and ID-team comments;
- preserves source citations and field-data traceability;
- documents residual uncertainty and remaining Forest Service decisions;
- final PDF starts with a valid '%PDF-' header.

SOURCES CITED

- 36 CFR 220.4, Forest Service NEPA general requirements: https://ecfr.io/Title-36/Section-220.4
- 36 CFR 220.7, Environmental assessment and decision notice: https://ecfr.io/Title-36/Section-220.7
- USDA Forest Service Manual, Region 1 Supplement, FSM 2550 Soil Management.

- Custer Gallatin National Forest Land Management Plan:
  https://www.fs.usda.gov/r01/custergallatin/planning/forest-plan/custer-gallatin-land-management-plan-forest-plan-revision
- Forest Service Soil Quality Monitoring resources:
  https://www.fs.usda.gov/soils/monitoring.shtml
- Forest Service Soil-Disturbance Field Guide:
  https://www.fs.usda.gov/t-d/php/library_card.php?p_num=0819+1815P
- NRCS Web Soil Survey and SSURGO:
  https://www.nrcs.usda.gov/resources/data-and-reports/web-soil-survey
- Forest Service National Best Management Practices Program:
  https://www.fs.usda.gov/naturalresources/watershed/bmp.shtml
- East Crazy Inspiration Divide Final Environmental Assessment, January 2025.
- East Crazies Plan Consistency Table.
- East Crazies Roads, Trails, and Access Report.
- East Crazies Wetlands Report.
- Post Exchange Federal Parcels and Restrictions.
- South Plateau Landscape Area Treatment Project, ‘Soils Report Revised_Anderson 2023.pdf‘, used as
  a Custer Gallatin soils report structure model.